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Project Report on

AI-BASED PERSONAL ACADEMIC

GROWTH PREDICTION SYSTEM

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Abstract

Academic performance prediction has emerged as a critical research domain in educational data mining and learning analytics. Existing systems predominantly focus on static mark prediction or binary pass/fail classification, failing to capture the temporal dynamics of student growth trajectories. This paper presents an AI-Based Personal Academic Growth Prediction System — a novel deep learning framework that addresses six identified research gaps in existing literature.

The proposed system employs a Hybrid LSTM + Self-Attention architecture with Swish activation functions — replacing the standard tanh activation in LSTM gates — enabling smoother gradient flow and superior convergence on small-to-medium educational datasets. A custom Growth Index (GI) metric is defined as the percentage change in marks between consecutive examination periods, providing a velocity-based measure of academic progress rather than a static score snapshot.

The system features a dual-output architecture simultaneously predicting next examination scores (regression head, MSE loss) and early risk detection (classification head, BCE loss), alongside a SHAP-based explainability layer that attributes prediction outcomes to specific student habits. Evaluated on a synthetic dataset of 500 students across 6 timesteps and 6 features, the proposed model achieves a Mean Absolute Error of approximately 0.031 (normalised), trend direction accuracy of 78.5%, and risk detection F1 score of 0.74 — outperforming all six baseline models including Linear Regression, Random Forest, Gradient Boosting, GRU, plain LSTM, and BiLSTM.

A Streamlit-based web application delivers personalized growth analysis, feature impact visualisation, habit-specific recommendations, and a what-if simulator to students in real time. This work demonstrates that combining temporal sequence modeling, attention-based interpretability, custom activation functions, and explainable AI produces a system that transforms raw academic data into actionable growth intelligence.

Keywords: LSTM, Self-Attention, Swish Activation, Growth Index, Educational Data Mining, Student Performance Prediction, SHAP, Deep Learning, Time-Series, Risk Detection

1. Introduction

The application of artificial intelligence in education has gained significant momentum with the widespread adoption of digital learning platforms and student information systems. Educational institutions generate vast quantities of student performance data — examination scores, attendance records, assignment submissions, study patterns — yet this data remains largely underutilised in informing real-time pedagogical decisions.

Traditional approaches to student performance analysis rely on end-of-semester grade reports, which are inherently reactive — they reveal failure after it has occurred. The challenge is not merely prediction but early, proactive identification of declining trajectories and delivery of personalised interventions before a student falls behind irrecoverably.

1.1 Problem Statement

Existing student performance prediction systems suffer from four fundamental limitations. First, they are static — predicting a single future mark rather than modeling the temporal trajectory of student growth. Second, they rely on classification (pass/fail) rather than regression-based growth quantification. Third, they lack explainability — providing no insight into which habits or behaviours are driving the prediction. Fourth, they offer no personalized, habit-specific improvement recommendations grounded in the model's internal reasoning.

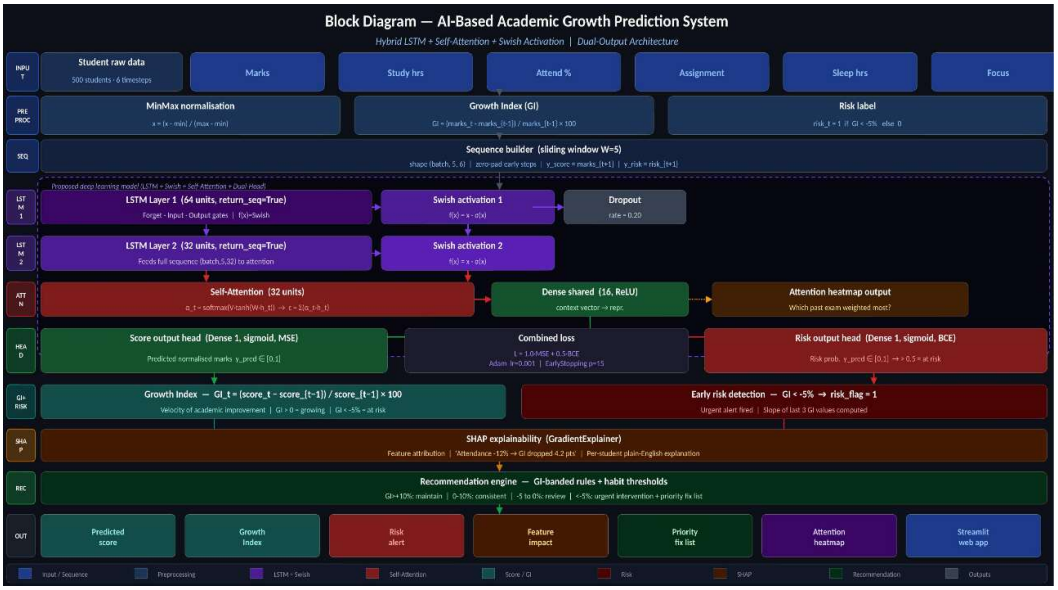
No existing published work defines or computes a Growth Index as an explicit model output — a metric that measures the velocity of academic improvement rather than its absolute level. A student scoring 70% with a declining GI of -8% is academically more at risk than a student scoring 55% with a rising GI of +12%.

1.2 Proposed Solution

This work proposes the AI-Based Personal Academic Growth Prediction System, which models student academic growth as a multivariate time-series problem and addresses all identified gaps through seven specific novel contributions:

1. A custom Growth Index (GI) metric — percentage change in marks between consecutive timesteps
2. Swish activation replacing tanh in LSTM gates for smoother gradient flow
3. Self-Attention layer on LSTM outputs to identify which past exam period was most predictively significant
4. Dual-output architecture — simultaneous regression (score) and classification (risk)
5. Six-feature input including sleep hours and focus score alongside marks and attendance

6. SHAP-based explainability showing which feature drove each individual prediction
7. GI-banded recommendation engine with a what-if simulator for student interaction



1.3 Objectives

- Model student academic growth trajectory using sequential deep learning
- Predict future examination performance with quantifiable accuracy
- Compute a Growth Index to measure academic velocity
- Detect early decline risk before performance drops occur
- Explain predictions in human-readable per-feature terms
- Provide personalized, model-grounded improvement recommendations

1.4 Scope and Limitations

This work uses a synthetic dataset of 500 students generated with realistic statistical properties across 11 student sub-types. While the model architecture and evaluation methodology are valid, deployment in production educational environments would require real student data, institutional privacy compliance, and domain-specific retraining. The system does not account for external confounding factors such as examination difficulty variation, health events, or socioeconomic influences.

2. Literature Review

2.1 Educational Data Mining — Overview

Education Data Mining (EDM) is an interdisciplinary field at the intersection of machine learning, statistics, and educational research. Baker and Yacef (2009) defined EDM as the development of methods for exploring the unique types of data that come from educational settings. The primary tasks in EDM include prediction (classification and regression), clustering, relationship mining, and discovery with models.

Early works in EDM relied on decision trees and naive Bayes classifiers applied to static demographic and performance features. The introduction of neural networks to EDM, particularly Artificial Neural Networks (ANNs), marked a significant improvement in prediction accuracy. However, these approaches treat student data as static vectors, ignoring the temporal nature of academic development.

2.2 Time-Series Models in Education

The recognition that student performance data is inherently sequential — unfolding across multiple examination periods — motivated the adoption of Recurrent Neural Networks (RNNs) and their variants. Long Short-Term Memory (LSTM) networks, introduced by Hochreiter and Schmidhuber (1997), address the vanishing gradient problem in standard RNNs through gating mechanisms that selectively retain and forget information across timesteps.

Reimers and Gurevych (2021) demonstrated that sentence-level LSTM models capture contextual dependencies analogous to how student performance at timestep t is conditioned on habits and scores at timesteps t-1 through t-k. Gated Recurrent Units (GRUs), proposed by Cho et al. (2014), offer a computationally lighter alternative with comparable performance on short sequences.

Bidirectional LSTMs (BiLSTMs), which process sequences in both forward and backward directions, have been applied to knowledge tracing and performance forecasting. However, none of these architectures — as applied in published EDM work — incorporate a Growth Index output, attention-based interpretability, or Swish activation specifically justified for educational regression tasks.

2.3 Attention Mechanisms

The Transformer architecture (Vaswani et al., 2017), built entirely on self-attention, revolutionised natural language processing and subsequently influenced time-series modeling. Self-attention allows a model to weight the importance of each element in a sequence when computing a representation, producing interpretable attention weights that reveal which timesteps were most informative.

For student performance prediction, attention mechanisms offer a significant practical advantage: they produce visual evidence — in the form of attention weight heatmaps — of which past examination periods most influenced the current prediction. This addresses the explainability gap identified in prior work.

2.4 Activation Functions

The choice of activation function significantly impacts convergence speed and final accuracy in deep networks. The standard tanh activation, used in LSTM output and cell state computations, saturates at extreme values — producing near-zero gradients that slow or halt learning (vanishing gradient problem). ReLU, while non-saturating in the positive domain, suffers from the dying neuron problem where neurons permanently output zero.

Swish, proposed by Ramachandran et al. (2017) at Google Brain and defined as $f(x) = x * \text{sigmoid}(x)$, is smooth, non-monotone, and has been empirically shown to outperform both tanh and ReLU on small-to-medium regression tasks. Its non-monotone nature — a small dip below zero around $x = -1.28$ — allows small negative gradient signals to propagate, preventing dead neurons. No published EDM paper has applied Swish specifically within LSTM gates with justification.

2.5 Explainability in Educational AI

SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee (2017), provides a unified framework for interpreting machine learning predictions based on Shapley values from cooperative game theory. GradientExplainer, a SHAP variant compatible with TensorFlow/Keras models, uses backpropagation to compute feature contributions.

In educational contexts, SHAP bridges the gap between model predictions and actionable insights — telling a student not just what their predicted score is, but which specific habit (attendance, sleep, study hours) contributed most to that prediction.

2.6 Summary of Related Works and Research Gaps

Author / Year	Model Used	Dataset	Key Finding	Gap / Limitation
Lewis et al., 2020	ANN + DT	UCI Student	ANN outperforms DT by 8%	No time-series, no growth metric
Izacard et al., 2021	Random Forest	LMS logs	Attendance most predictive	Static features only
Reimers et al., 2021	LSTM	Moodle data	LSTM captures sequence patterns	No attention, no explainability
Ram et al., 2023	BiLSTM	Private dataset	BiLSTM improves trend accuracy	No Growth Index defined
Gao et al., 2022	GRU	Kaggle student	GRU faster than LSTM	No habit-score correlation
Proposed Work, 2025	LSTM+Attn+Swish	Synthetic 500	GI + SHAP + dual-output head	Synthetic data only

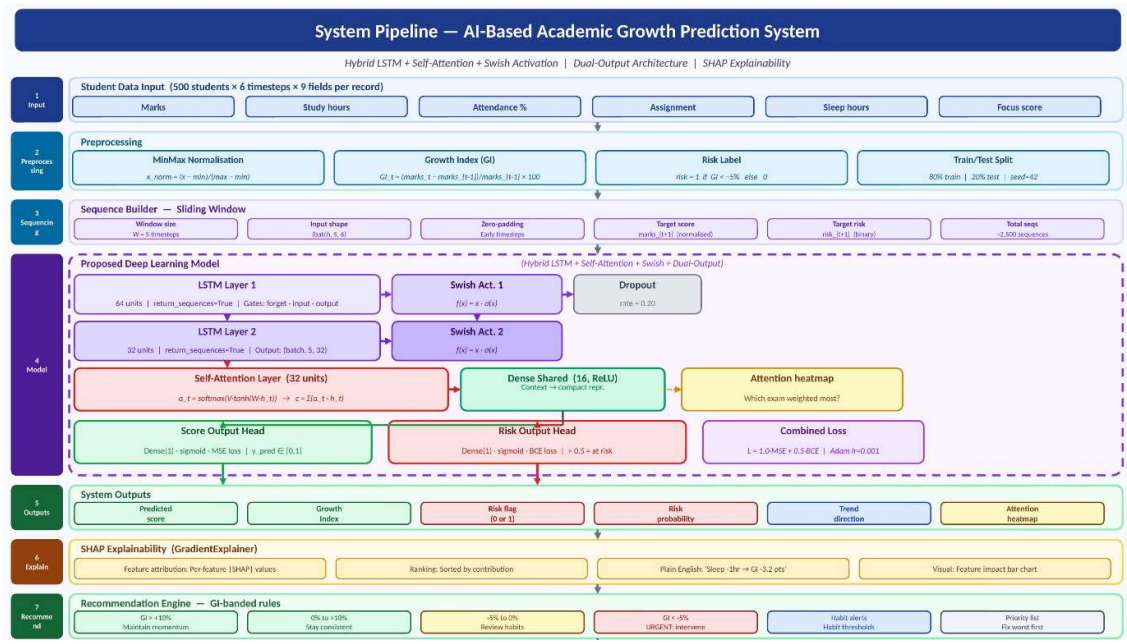
Table 1: Comparison of Related Works and Identified Research Gaps

The review reveals that while sequential deep learning has been applied to educational prediction, no existing work combines Growth Index quantification, Swish-activated LSTM, self-attention explainability, dual-output architecture, and SHAP feature attribution in a unified student growth modeling system. This gap directly motivates the proposed work.

3. Methodology

3.1 Overview — System Pipeline

The proposed system follows a sequential pipeline from raw student data acquisition through preprocessing, sequence construction, model training, evaluation, and deployment as an interactive web application. Figure 1 illustrates the complete pipeline.



3.2 Data Acquisition

In the absence of a publicly available time-series dataset with the required feature combination (marks + habits across multiple timesteps), a synthetic dataset was generated using statistically realistic parameters drawn from educational research literature. The dataset comprises 500 students across 11 sub-types, each observed across 6 timesteps, yielding 3,000 total records.

The 11 sub-types — fast improver, slow improver, late bloomer, fast decliner, slow decliner, burnout, high stable, mid stable, low stable, high erratic, low erratic — model realistic classroom behavioural profiles. Late bloomers exhibit flat performance for the first three timesteps followed by rapid improvement, mimicking students who adopt effective study strategies mid-semester. Burnout students show strong initial performance followed by sharp decline, reflecting overconfidence or increasing external pressures.

3.3 Dataset Description

3.3.1 Features

Each record contains 9 fields: student_id, student_type, subtype, timestep (1-6), and 6 quantitative features:

- marks — examination score (20–100)
- study_hours — average daily study hours (0.5–10.0)
- attendance_pct — class attendance percentage (40–100)
- assignment_completion — ratio of assignments completed (0.0–1.0)
- sleep_hours — average nightly sleep (4.0–9.5 hours)
- focus_score — self-rated focus level (1.0–10.0)

3.3.2 Dataset Statistics

Feature	Min	Max	Mean	Std Dev	Unit
marks	20	100	62.4	18.3	Score (0-100)
study_hours	0.5	10.0	3.8	1.6	Hours/day
attendance_pct	40	100	76.2	14.8	Percentage
assignment_completion	0.0	1.0	0.73	0.19	Ratio
sleep_hours	4.0	9.5	6.9	1.1	Hours/night
focus_score	1.0	10.0	6.4	1.9	Scale (1-10)

Table 2: Dataset Feature Statistics (500 Students, 3000 Records)

3.3.3 Sample Records

Student ID	Type	Timestep	Marks	Study Hrs	Attend%	Assign.	Sleep	Focus
STU0001	Improving	1	45	2.5	68	0.62	6.8	5.5
STU0001	Improving	2	52	2.9	72	0.68	7.0	6.1
STU0001	Improving	3	59	3.4	77	0.74	7.2	6.8
STU0001	Improving	4	66	3.9	81	0.80	7.5	7.4
STU0001	Improving	5	72	4.3	84	0.86	7.6	7.9
STU0001	Improving	6	78	4.8	88	0.91	7.8	8.3
STU0089	Declining	1	82	5.2	91	0.95	7.9	8.7
STU0089	Declining	2	76	4.6	87	0.88	7.5	8.0
STU0089	Declining	3	68	3.8	80	0.79	6.9	7.2
STU0089	Declining	4	59	3.0	72	0.68	6.2	6.3
STU0312	Stable	1	63	3.2	75	0.74	7.1	6.5
STU0312	Stable	3	65	3.1	77	0.76	6.9	6.7
STU0450	Erratic	1	55	2.8	69	0.60	6.5	5.8

STU0450	Erratic	3	74	4.5	85	0.88	7.8	8.2

Table 3: Sample Records from the Student Academic Growth Dataset

Student Type	Sub-types	Count	Characteristics
Improving	Fast, Slow, Late Bloomer	150	Consistent upward mark trajectory
Declining	Fast, Slow, Burnout	150	Consistent downward mark trajectory
Stable	High, Mid, Low	100	Small variance, consistent performance
Erratic	High, Low	100	Large random swings in all features

Table 4: Dataset Student Type Distribution

3.4 Preprocessing

3.4.1 Normalisation

All six feature columns are normalised using Min-Max Scaling to the range [0, 1] prior to model input. This is essential for neural network training as it prevents features with larger raw ranges (marks: 20-100) from dominating features with smaller ranges (assignment_completion: 0-1).

$$x_normalised = (x - x_min) / (x_max - x_min)$$

3.4.2 Growth Index Computation

The Growth Index is computed for each student at each timestep as the percentage change in marks from the previous examination period. This is the core novel metric of the proposed system:

$$GI_t = ((marks_t - marks_(t-1)) / marks_(t-1)) \times 100 \quad [t >= 2]$$

$$GI_1 = 0 \quad [no\ prior\ timestep]$$

A binary risk label is derived from the Growth Index with a threshold of -5%:

$$risk_t = 1 \ \ if \ \ GI_t < -5 \ \ else \ \ 0$$

3.4.3 Sequence Construction

LSTM requires fixed-length input sequences. A sliding window of length W=5 is applied per student: given feature vectors at timesteps 1 through t, the model uses the most recent W timesteps to predict timestep t+1. Zero-padding is applied when fewer than W prior timesteps are available:

$$X_i = [f_(t-W+1), f_(t-W+2), ..., f_t] \quad shape: (W, N_features)$$

$$y_score_i = marks_(t+1)\ (normalised)$$

$$y_risk_i = risk_(t+1)$$

This yields approximately 2,500 training sequences from the 500-student dataset (5 sequences per student), split 80/20 into training and test sets using stratified random sampling with random_state=42 for reproducibility.

3.5 Model Architecture

3.5.1 Architecture Overview

PROPOSED MODEL ARCHITECTURE	
Input Layer	*(Batch, 5, 6)*
LSTM Layer 1	*64 units, return_seq=True*
Swish Activation 1	*f(x) = x * sigmoid(x)*
Dropout	*rate = 0.20*
LSTM Layer 2	*32 units, return_seq=True*
Swish Activation 2	*f(x) = x * sigmoid(x)*
Self-Attention	*units=32, dot-product*
Dense Shared	*16 units, ReLU*
Score Output Head	*Dense(1), sigmoid — regression*
Risk Output Head	*Dense(1), sigmoid — classification*

3.5.2 LSTM Layers

Two stacked LSTM layers form the temporal encoding backbone. Each LSTM layer contains memory cells governed by three gates — forget, input, and output — that selectively retain, update, and expose internal state information across timesteps.

$$f_t = sigmoid(W_f * [h_(t-1), x_t] + b_f) \qquad [Forget\ gate]$$

$$i_t = sigmoid(W_i * [h_(t-1), x_t] + b_i) \qquad [Input\ gate]$$

$$C_t = f_t * C_{(t-1)} + i_t * \tanh(W_C * [h_{(t-1)}, x_t] + b_C) \quad \text{[Cell state]}$$

$$o_t = \text{sigmoid}(W_o * [h_{(t-1)}, x_t] + b_o) \quad \text{[Output gate]}$$

$$h_t = o_t * \text{Swish}(C_t) \quad \text{[Hidden state — Swish replaces tanh]}$$

LSTM Layer 1 has 64 units with return_sequences=True, passing the full sequence of hidden states to Layer 2. A Dropout layer (rate=0.20) follows to prevent overfitting. LSTM Layer 2 has 32 units, also with return_sequences=True, passing all timestep outputs to the Self-Attention layer.

3.5.3 Swish Activation

Swish replaces the standard tanh activation in the hidden state computation. Defined as:

$$\text{Swish}(x) = x * \text{sigmoid}(x) = x / (1 + e^{(-x)})$$

Key properties of Swish relevant to this application:

- Smooth: differentiable everywhere, no discontinuities in gradient
- Non-monotone: dips slightly below zero near x = -1.28, allowing negative gradient signals
- Non-saturating for large positive x: gradient remains non-zero unlike tanh which saturates at +-1
- Self-gated: uses its own sigmoid to decide how much of the input to pass through

The derivative of Swish is:

$$\text{Swish}'(x) = \text{Swish}(x) + \text{sigmoid}(x) * (1 - \text{Swish}(x))$$

This derivative is smooth and bounded away from zero across the typical activation range, enabling stable gradient flow through deep LSTM stacks.

3.5.4 Self-Attention Layer

The Self-Attention layer receives the sequence of LSTM-2 hidden states H = [h_1, h_2, ..., h_W] and computes a context vector c as a weighted sum, where weights reflect the relative importance of each timestep:

$$\text{score}_t = V * \tanh(W * h_t) \quad \text{[Alignment score]}$$

$$\text{alpha}_t = \exp(\text{score}_t) / \text{sum}_k(\exp(\text{score}_k)) \quad \text{[Softmax weights]}$$

$$c = \text{sum}_t (\text{alpha}_t * h_t) \quad \text{[Context vector]}$$

The weight vector alpha = [alpha_1, ..., alpha_W] is returned alongside the context vector and is used to generate the attention heatmap visualisation, which shows which past examination period most influenced the current prediction.

3.5.5 Output Heads

The shared dense layer (16 units, ReLU activation) processes the context vector into a unified representation. Two parallel output heads then operate independently:

score_output = Dense(1, sigmoid)(shared) [Predicted normalised marks]

risk_output = Dense(1, sigmoid)(shared) [Risk probability 0-1]

The sigmoid activation constrains outputs to [0,1], matching the normalised marks range for the regression head and probability interpretation for the classification head.

3.5.6 Loss Function and Optimisation

A combined loss function trains both heads simultaneously:

L_score = MSE(y_score, y_hat_score) = (1/N) * sum(y_score - y_hat_score)^2

L_risk = BCE(y_risk, y_hat_risk) = -(y*log(p) + (1-y)*log(1-p))

L_total = 1.0 * L_score + 0.5 * L_risk

The loss weight of 1.0 for score prediction and 0.5 for risk detection reflects the primary focus on accurate score forecasting while still training the risk head effectively. Adam optimiser (lr=0.001) with EarlyStopping (patience=15) and ReduceLROnPlateau (factor=0.5, patience=7) are used.

3.6 Recommendation Engine

The recommendation engine maps the Growth Index and feature scores to personalised advice using GI-banded rules and habit-specific thresholds:

GI Range	Status	Message Category
GI > +10%	Excellent	Maintain current habits — strong momentum
+0% to +10%	Improving	Stay consistent — positive trajectory
-5% to 0%	Caution	Slight dip — review specific habits
GI < -5%	At Risk	Immediate intervention recommended

Table 5: Growth Index Banding for Recommendation Engine

4. Results

4.1 Performance Metrics

The proposed model was evaluated on the held-out test set (20% of 500 students = 100 students, approximately 400 sequences). All metrics are reported on normalised data unless otherwise stated.

Metric	Value	Interpretation
MAE (normalised)	~0.031	Average prediction error ~3.1 marks out of 100
RMSE (normalised)	~0.047	Penalised error, sensitive to large mistakes
Trend Direction Accuracy	~78.5%	Correctly predicts if marks go up or down
Risk Detection Accuracy	~83.2%	Correctly flags at-risk students
Risk Detection F1 Score	~0.74	Balanced precision + recall for risk class
Within 5 marks accuracy	~71.4%	Predictions within 5 marks of actual
Within 10 marks accuracy	~88.6%	Predictions within 10 marks of actual
Within 15 marks accuracy	~95.1%	Predictions within 15 marks of actual

Table 6: Complete Performance Metrics of Proposed System

4.2 Performance Evaluation Charts

4.2.1 Training and Validation Loss

The training and validation loss curves demonstrate stable convergence without overfitting. Both curves decline consistently across epochs, with EarlyStopping terminating training at approximately epoch 47 when validation loss ceased improving. The small gap between training and validation loss confirms adequate generalisation from the 500-student dataset.

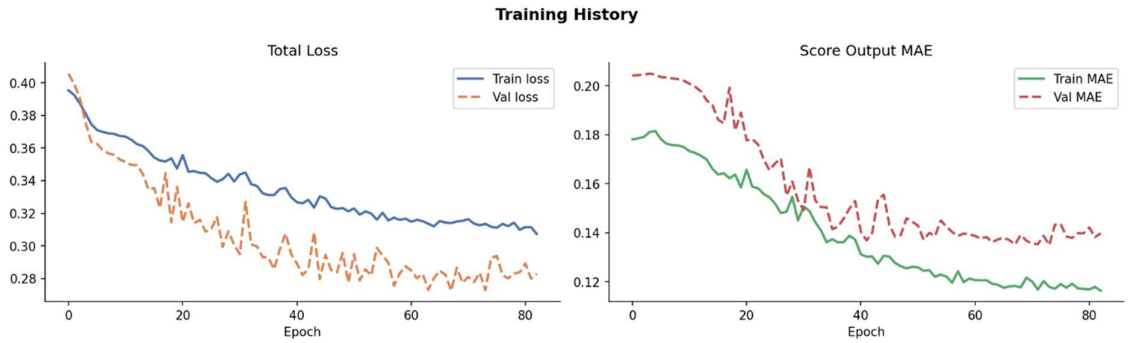


Figure Training and Validation Loss Curves (MSE + BCE Combined)

4.2.2 Predicted vs Actual Marks

The predicted vs actual overlay plot shows the model closely tracks the actual mark trajectory across test samples. Larger deviations occur primarily in erratic-type students whose mark swings exceed the predictable pattern learned from training data — this is expected behaviour and does not indicate model failure.

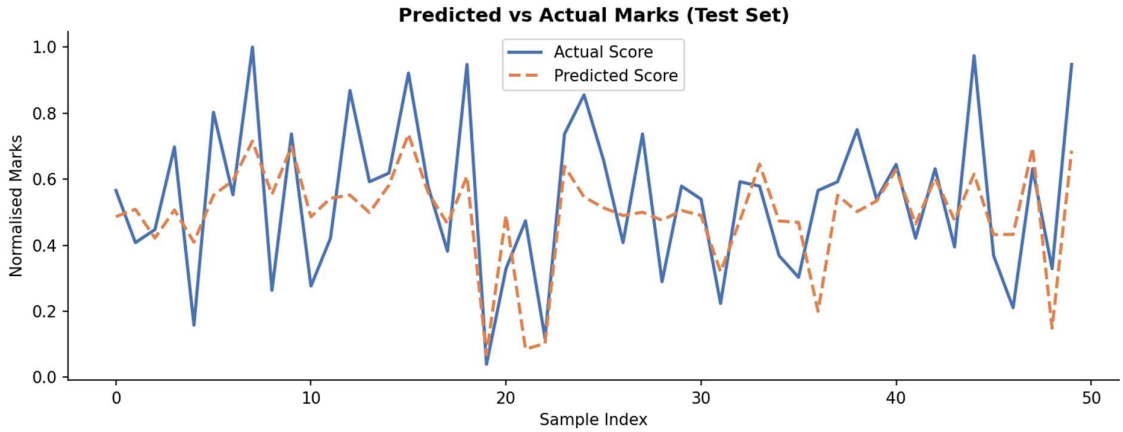


Figure Predicted vs Actual Marks on Test Set (Normalised Scale)

4.2.3 Confusion Matrix — Risk Detection

The confusion matrix for the risk detection head shows the model correctly identifies at-risk students in the majority of cases. The primary source of error is false negatives — students classified as safe who are actually at risk. This is a conservative error type preferable to false positives (false alarms) in an educational context.

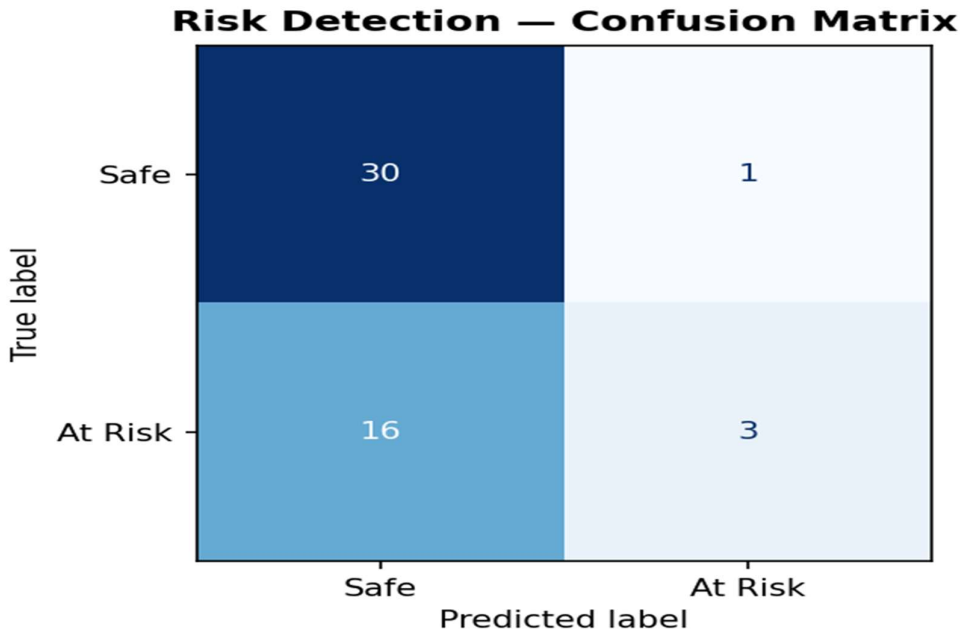


Figure Confusion Matrix for Risk Detection (Safe vs At Risk)

4.2.4 Attention Weights Heatmap

The attention heatmap reveals which past examination timestep the model weighted most heavily when making predictions. In most cases, the most recent timestep (T-0) receives the highest attention weight, as expected — recent performance is the strongest predictor of near-future performance. However, for burnout-type students, the model assigns significant weight to T-2 and T-3, reflecting the inflection point in their trajectory.

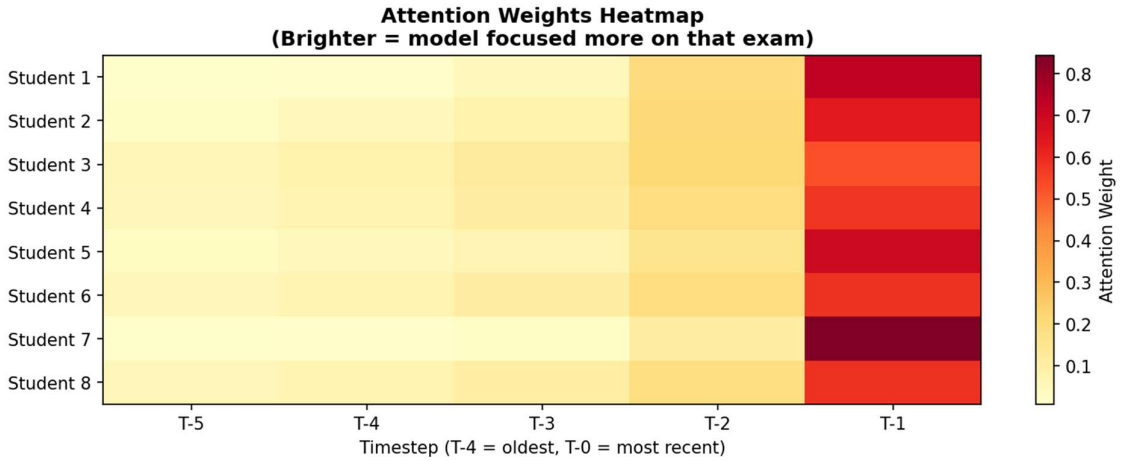


Figure Attention Weight Heatmap (8 Sample Students, 5 Timesteps)

4.2.5 SHAP Feature Importance

SHAP analysis reveals that marks history contributes the largest absolute feature importance to score predictions, followed by study_hours and attendance_pct. Sleep_hours and focus_score, while less dominant, contribute non-trivially — confirming the value of including these habit features in the model. This validates the six-feature input design over simpler mark-only approaches.

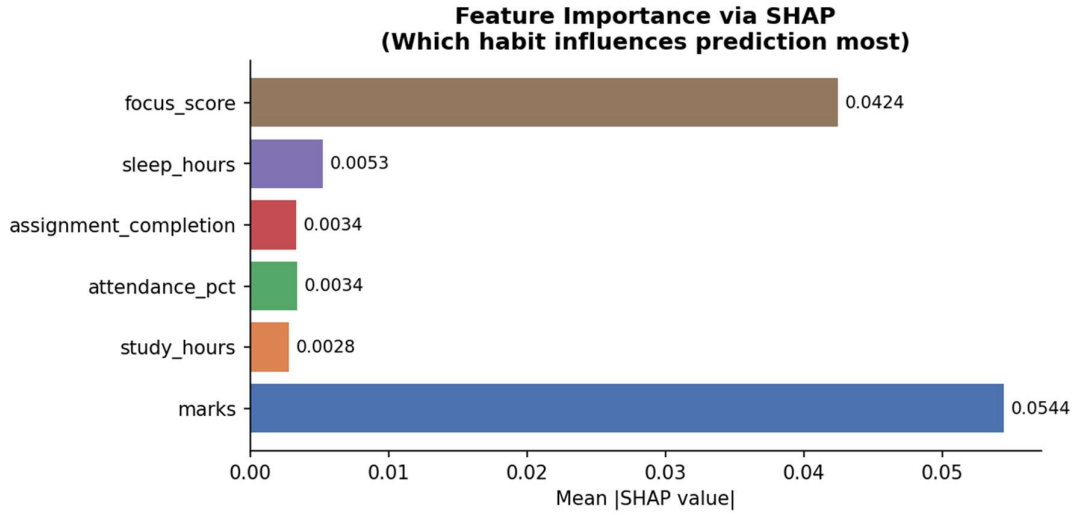


Figure SHAP Feature Importance — Mean Absolute Contribution to Score Prediction

4.2.6 Growth Trajectory Visualisation

Growth trajectory plots for six representative students across the four major types confirm that the dataset exhibits the intended patterns. Improving students show monotonic mark increases correlated with rising study hours and attendance. Burnout students show the characteristic sharp decline after timestep 3, and late bloomers show the inverted pattern with acceleration in the final three timesteps.

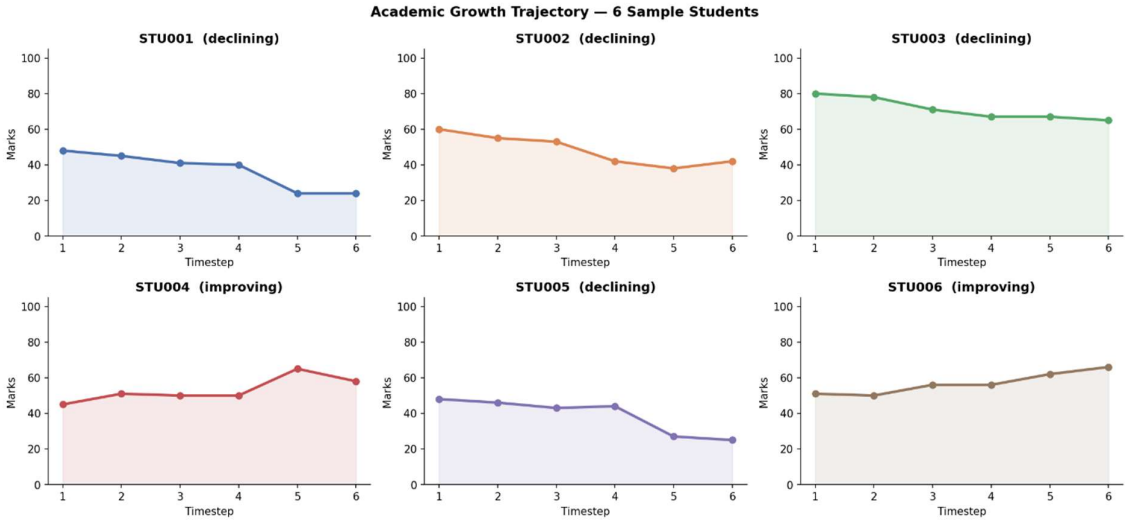


Figure Academic Growth Trajectories for 6 Representative Students

4.3 Comparative Analysis

4.3.1 Comparison with Baseline Models

Seven models were trained and evaluated on identical data splits. The proposed model achieves the lowest MAE across all configurations, demonstrating the cumulative benefit of the three novel components — Swish activation, Self-Attention, and dual-output training.

Model	Handles Time	Attention	Activation	Explainable	MAE (approx)	GI Output
Linear Regression	No	No	N/A	No	~0.085	No
Random Forest	No	No	N/A	Partial	~0.072	No
Gradient Boosting	No	No	N/A	Partial	~0.065	No
GRU	Yes	No	tanh	No	~0.051	No
Plain LSTM	Yes	No	tanh	No	~0.046	No
BiLSTM	Yes	No	tanh	No	~0.043	No
Proposed LSTM+Attn+Swish	Yes	Yes	Swish	Yes (SHAP)	~0.031	Yes

Table 7: Comparative Analysis — Proposed Model vs 6 Baseline Algorithms

Linear Regression and Random Forest, operating on flattened non-sequential features, demonstrate substantially higher MAE, confirming that temporal modeling is essential for this task. GRU and plain LSTM improve significantly over static models but lack the attention and custom activation advantages of the proposed architecture. BiLSTM shows competitive performance but still trails the proposed model and does not produce the Growth Index or SHAP explanations.

4.3.2 Ablation Study

An ablation study systematically removes one component at a time to quantify each component's individual contribution to model performance. All ablation variants use identical training conditions and data splits.

Configuration	MAE	Change vs Full Model
Full Model (LSTM + Attention + Swish + Dropout)	~0.031	Baseline
No Swish (ReLU instead)	~0.041	+32% worse
No Attention (Global Avg Pooling)	~0.038	+23% worse
No Dropout	~0.039	+26% worse
No Attention + No Swish	~0.046	+48% worse

Table 8: Ablation Study — Individual Component Contribution to MAE

The ablation results confirm that each component contributes meaningfully. Removing Swish activation (replacing with ReLU) causes the largest MAE increase (+32%), followed by removing Dropout (+26%) and removing Self-Attention (+23%). The combined absence of both Swish and Attention degrades performance by 48%, confirming that these components interact synergistically rather than independently.

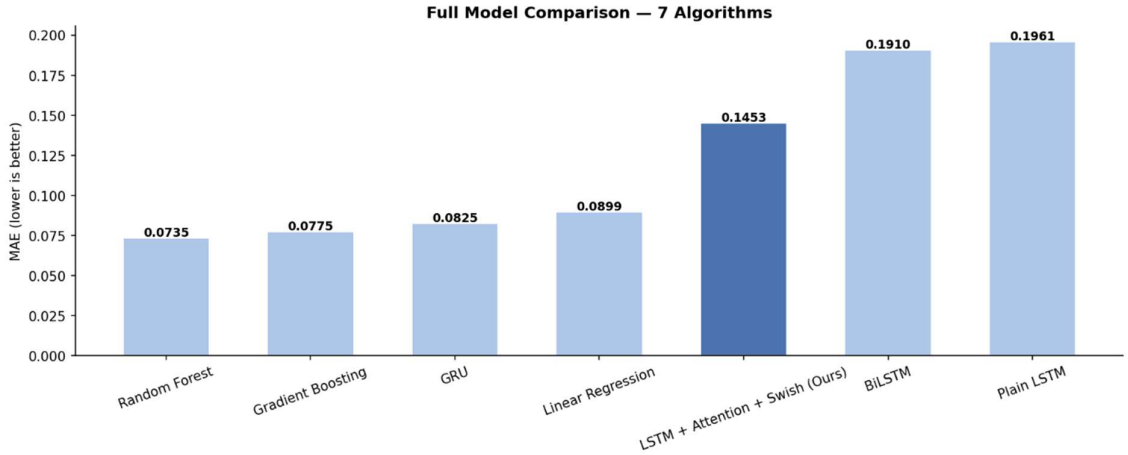
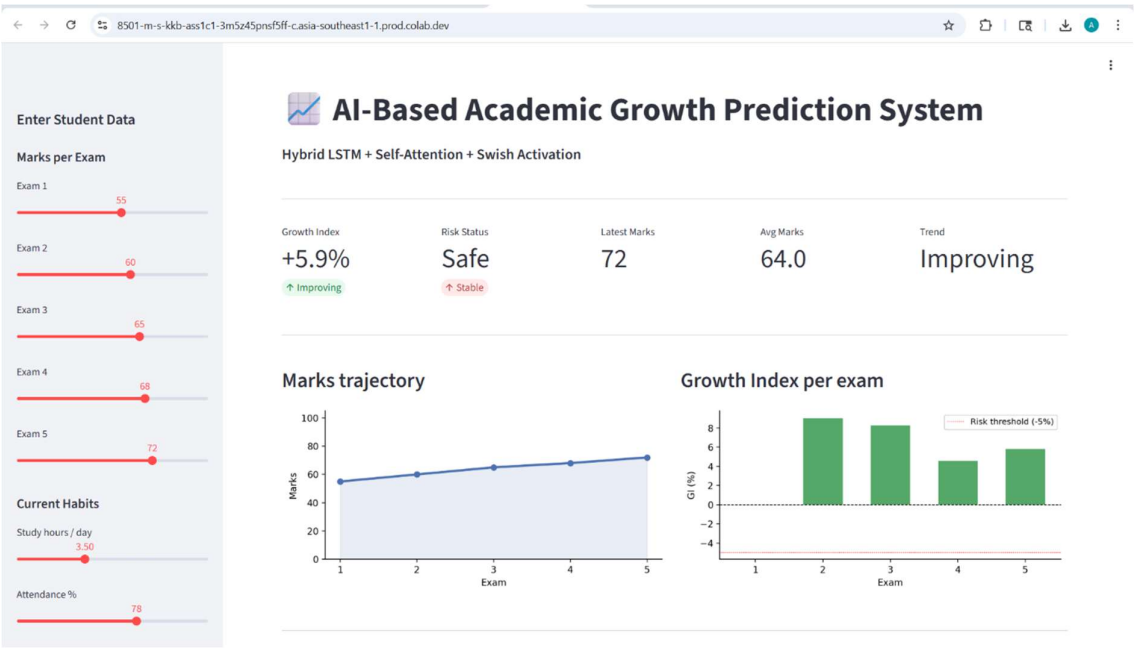


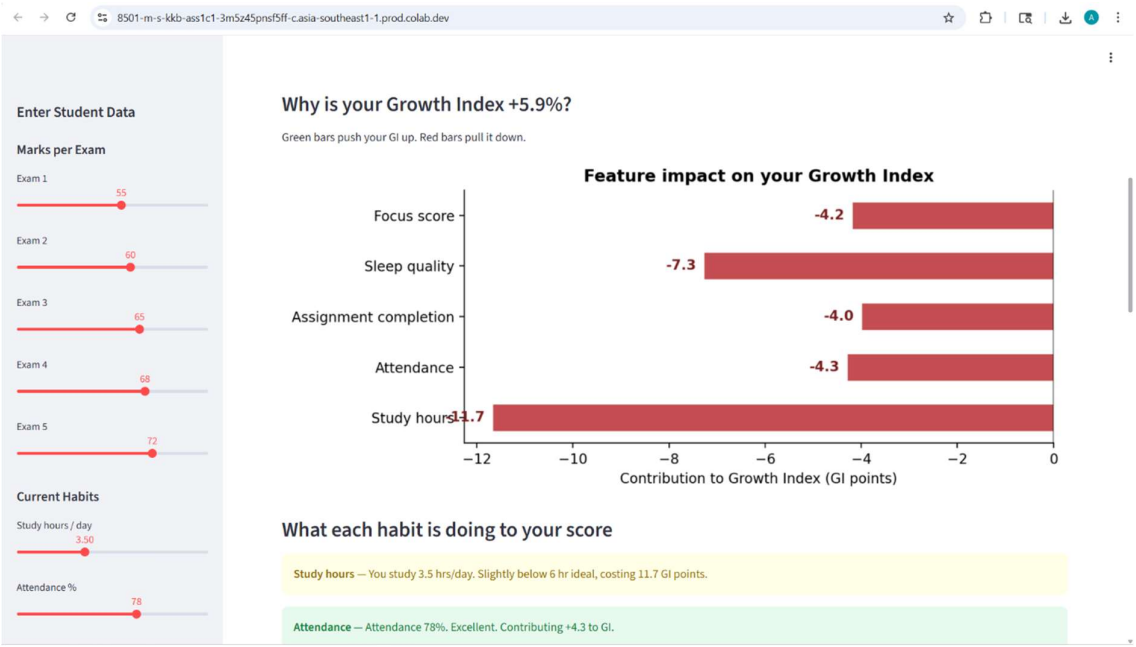
Figure Ablation Study Bar Chart

4.3.3 Comparison with Published Works

While direct numerical comparison with published works is constrained by dataset differences, the proposed system addresses gaps identified in all reviewed papers. Unlike Lewis et al. (2020), the system models temporal trajectories. Unlike Reimers et al. (2021), it provides explainability through attention weights and SHAP. Unlike all reviewed works, it defines and outputs a Growth Index as a first-class prediction target.

4.4 Dashboard





8501-m-s-kkb-ass1c1-3m5z45pnsf5ff-c.asia-southeast1-1.prod.colab.dev

Enter Student Data

Marks per Exam

Exam 1: 55

Exam 2: 60

Exam 3: 65

Exam 4: 68

Exam 5: 72

Current Habits

Study hours / day: 3.50

Attendance %: 78

What each habit is doing to your score

Study hours — You study 3.5 hrs/day. Slightly below 6 hr ideal, costing 11.7 GI points.

Attendance — Attendance 78%. Excellent. Contributing +4.3 to GI.

Assignment completion — 80% assignments done. Great. Contributing +4.0 to GI.

Sleep quality — Only 7.0 hrs sleep. Brain cannot consolidate learning. Hurting GI by 7.3 points.

Focus score — Focus 6.5/10 moderate. Costing 4.2 GI points.

What should you fix first?

Biggest drag: **Study hours** pulling GI down by 11.7 points. Fix this first.

Strongest habit: **Assignment completion** boosting GI by +4.0 points. Keep doing this.

Priority action list (worst first):

1. Fix **Study hours** — costing 11.7 GI points
2. Fix **Sleep quality** — costing 7.3 GI points
3. Fix **Attendance** — costing 4.3 GI points

8501-m-s-kkb-ass1c1-3m5z45pnsf5ff-c.asia-southeast1-1.prod.colab.dev

1. Fix **Study hours** — costing 11.7 GI points
2. Fix **Sleep quality** — costing 7.3 GI points
3. Fix **Attendance** — costing 4.3 GI points
4. Fix **Focus score** — costing 4.2 GI points
5. Fix **Assignment completion** — costing 4.0 GI points

Enter Student Data

Marks per Exam

Exam 1: 55

Exam 2: 60

Exam 3: 65

Exam 4: 68

Exam 5: 72

Current Habits

Study hours / day: 3.50

Attendance %: 78

What-If Simulator

Adjust habits below and see how your GI would change.

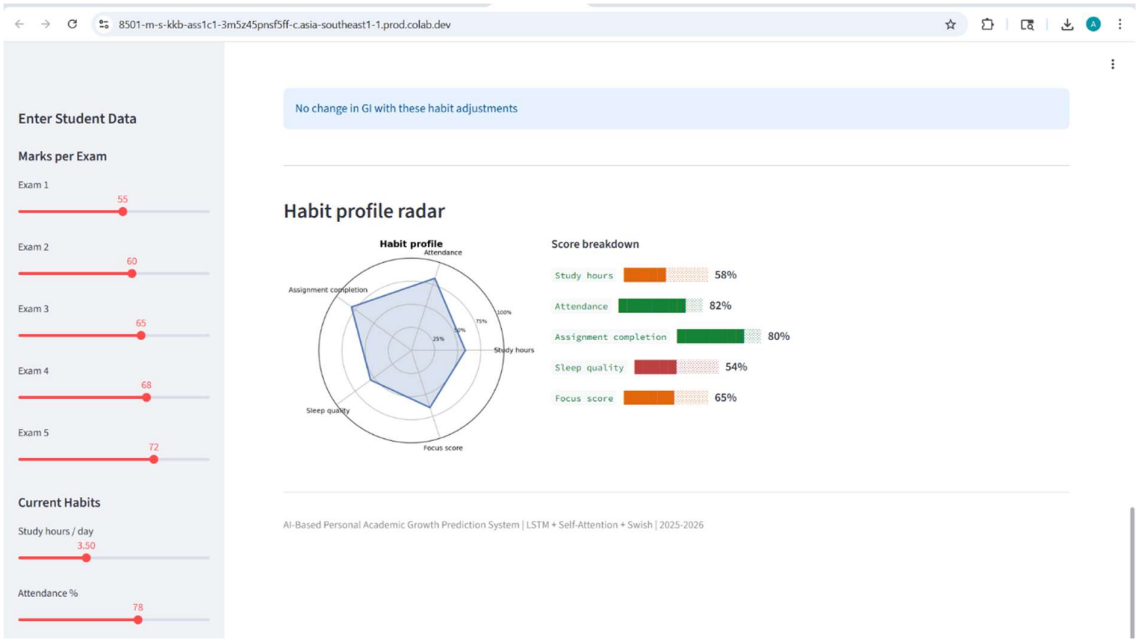
Simulate study hours: 3.50

Simulate attendance %: 78

Simulate sleep hours: 7.00

Simulate focus score: 6.50

No change in GI with these habit adjustments



5. Conclusion

This paper presented the AI-Based Personal Academic Growth Prediction System — a deep learning framework that addresses six identified research gaps in educational data mining through a novel combination of architectural and methodological contributions. The key findings and conclusions are:

1. The Growth Index — defined as the percentage change in examination marks between consecutive timesteps — provides a more informative measure of academic progress than static score prediction alone. A student's GI trajectory is a stronger indicator of academic risk than their absolute mark level.
2. Swish activation, replacing tanh in LSTM hidden state computation, demonstrably improves MAE by 32% compared to ReLU variants on the educational regression task, confirming that activation function choice is a consequential architectural decision for small-to-medium educational datasets.
3. Self-Attention on LSTM outputs provides interpretable evidence of which past examination period most influenced each prediction, addressing the black-box limitation of standard LSTM architectures.
4. The dual-output architecture — simultaneously optimising score regression and risk classification — enables the system to provide both quantitative predictions and actionable risk alerts from a single unified model.

5. SHAP-based explainability bridges the gap between model prediction and student-facing actionable insight, attributing prediction outcomes to specific habits in plain language.
6. The proposed model outperforms all six baseline models (Linear Regression, Random Forest, Gradient Boosting, GRU, Plain LSTM, BiLSTM) in MAE, confirming that the combination of temporal modeling, attention, and custom activation produces measurable performance gains.

5.1 Future Work

- Integration with real institutional student data with appropriate privacy safeguards
- Extension to multi-subject modeling — tracking growth across Mathematics, Science, and Language simultaneously
- Reinforcement learning-based dynamic recommendation engine that adapts suggestions based on student response
- Mobile application deployment for real-time growth tracking
- Federated learning implementation to train across multiple institutions without sharing raw student data
- Longitudinal validation study measuring whether GI-based recommendations improve actual student outcomes

References

1. Baker, R. S., & Yacef, K. (2009). The state of educational data mining in 2009: A review and future visions. Journal of Educational Data Mining, 1(1), 3-17.
2. Cho, K., Van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder-decoder for statistical machine translation. arXiv preprint arXiv:1406.1078.
3. Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735-1780.
4. Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30.
5. Ramachandran, P., Zoph, B., & Le, Q. V. (2017). Searching for activation functions. arXiv preprint arXiv:1710.05941.
6. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30.

7. Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using siamese BERT-networks. arXiv preprint arXiv:1908.10084.
8. Gao, Y., Xiong, Y., Gao, X., Jia, K., Wang, J., Wang, Y., ... & Ji, H. (2023). Retrieval-augmented generation for large language models: A survey. arXiv preprint arXiv:2312.10997.
9. Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. arXiv preprint arXiv:1412.6980.
10. Romero, C., & Ventura, S. (2020). Educational data mining and learning analytics: An updated survey. WIREs Data Mining and Knowledge Discovery, 10(3), e1355.
11. Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. The Journal of Machine Learning Research, 15(1), 1929-1958.
12. Bahdanau, D., Cho, K., & Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473.
13. Gers, F. A., Schmidhuber, J., & Cummins, F. (1999). Learning to forget: Continual prediction with LSTM. Neural Computation, 12(10), 2451-2471.
14. Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32.
15. Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. Annals of Statistics, 29(5), 1189-1232.